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Grupa 341

SSI - Lab 5

1. sample1.js

a. Codul executa o functie obfuscata si afiseaza un mesaj in consola.

b. Am modificat codul pentru a afisa mesajul construit in loc sa il ruleze. Astfel, am reusit sa vad ce text era ascuns initial. De asemenea, am observat ca **eval()** era folosit pentru a executa un string generat dinamic, si am extras acel string pentru a-l afisa separat, fara sa rulez efectiv functia anonima.



c. Mesajul ascuns este : **`var ascuns = "Mesaj ascuns: 18367622009998665"`**

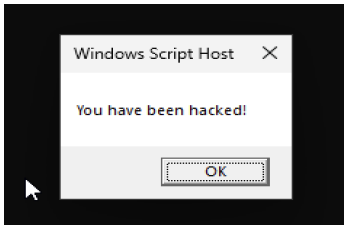
d. Codul este scris folosind o tehnica numita JSFuck, un mod de a construi programe JavaScript folosind doar caractere precum [], (), !, + si {}. Acest stil este adesea folosit pentru a obfusca codul, facandu-l greu de inteles la prima vedere.

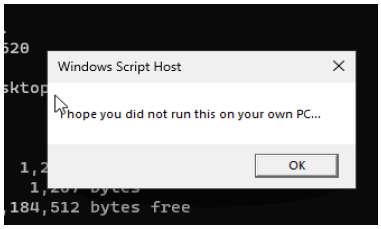
2. sample2.js

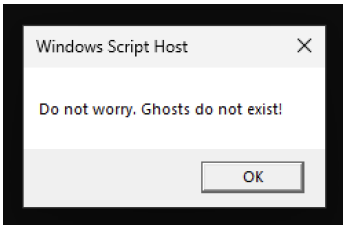
a. In momentul rularii pe masina virtuala, apar trei pop-up uri pe care le mentionez in primele linii de cod deobfuscat (2 din ele), iar cealalta din catch. Incearca totusi sa creeze un fisier ascuns (fmi.txt), folosind atributul 2 in cadrul functiei OpenTextFile.

```
WScript.Echo("You have been hacked!");
	WScript.Echo("I hope you did not run this on your own PC...");
	var f = "Facultatea";
```

```
var mi = "de Matematica si Informatica";
var unibuc = "Universitatea din Bucuresti";
var curs = "Curs Info anul 3";
var minciuna = "Acesta este un malware. Dispozitivul este compromis";
var adevar = "Stringul anterior este o minciuna";
try {
	var obj = new ActiveXObject("Scripting.FileSystemObject");
	var out = obj.OpenTextFile("./fmi.txt", 2, true, 0);
	out.WriteLine("Bun venit la acest laborator :)");
	out.Close();
	var fle = obj.GetFile("./fmi.txt");
	fle.attributes = 2
} catch (err) {
	WScript.Echo("Do not worry. Ghosts do not exist!")
}
```







b. Nu reprezinta un malware, intrucat nu cauzeaza daune sau pierderi de date, nici instructiuni daunatoare nu sunt rulate, si nici nu colecteaza sau transmite date. Doar se creeaza un fisier.

c.

3. sample3.js

a. Acest cod face acelasi lucru cu sample2.js. In schimb, se foloseste de caractere escape hexa pentru a ascunde textul real.

```
var _0x1d78 =
["\x59\x6F\x75\x20\x68\x61\x76\x65\x20\x62\x65\x65\x6E\x20\x68\x61\x63\x6B\x65\x64\x21",
"\x49\x20\x68\x6F\x70\x65\x20\x79\x6F\x75\x20\x64\x69\x64\x20\x6E\x6F\x74\x20\x72\x75\x6E\x20\
x74\x68\x69\x73\x20\x6F\x6E\x20\x79\x6F\x75\x72\x20\x6F\x77\x6E\x20\x50\x43\x2E\x2E\x2E",
"\x46\x61\x63\x75\x6C\x74\x61\x74\x65\x61",
"\x64\x65\x20\x4D\x61\x74\x65\x6D\x61\x74\x69\x63\x61\x20\x73\x69\x20\x49\x6E\x66\x6F\x72\x6D\
x61\x74\x69\x63\x61",
"\x55\x6E\x69\x76\x65\x72\x73\x69\x74\x61\x74\x65\x61\x20\x64\x69\x6E\x20\x42\x75\x63\x75\x72\
x65\x73\x74\x69", "\x43\x75\x72\x73\x20\x49\x6E\x66\x6F\x20\x61\x6E\x75\x6C\x20\x33",
"\x41\x63\x65\x73\x74\x61\x20\x65\x73\x74\x65\x20\x75\x6E\x20\x6D\x61\x6C\x77\x61\x72\x65\x2E\
x20\x44\x69\x73\x70\x6F\x7A\x69\x74\x69\x76\x75\x6C\x20\x65\x73\x74\x65\x20\x63\x6F\x6D\x70\x7
2\x6F\x6D\x69\x73",
"\x53\x74\x72\x69\x6E\x67\x75\x6C\x20\x61\x6E\x74\x65\x72\x69\x6F\x72\x20\x65\x73\x74\x65\x20\
x6F\x20\x6D\x69\x6E\x63\x69\x75\x6E\x61",
"\x53\x63\x72\x69\x70\x74\x69\x6E\x67\x2E\x46\x69\x6C\x65\x53\x79\x73\x74\x65\x6D\x4F\x62\x6A\
x65\x63\x74", "\x2E\x2F\x66\x6D\x69\x2E\x74\x78\x74",
"\x42\x75\x6E\x20\x76\x65\x6E\x69\x74\x20\x6C\x61\x20\x61\x63\x65\x73\x74\x20\x6C\x61\x62\x6F\
```

```
x72\x61\x74\x6F\x72\x20\x3A\x29", "\x61\x74\x74\x72\x69\x62\x75\x74\x65\x73",
"\x44\x6F\x20\x6E\x6F\x74\x20\x77\x6F\x72\x72\x79\x2E\x20\x47\x68\x6F\x73\x74\x73\x20\x64\x6F\
x20\x6E\x6F\x74\x20\x65\x78\x69\x73\x74\x21"];
```

```
WScript.Echo(_0x1d78[0]);
WScript.Echo(_0x1d78[1]);
var f = _0x1d78[2];
var mi = _0x1d78[3];
var unibuc = _0x1d78[4];
var curs = _0x1d78[5];
var minciuna = _0x1d78[6];
var adevar = _0x1d78[7];
try {
var obj = new ActiveXObject(_0x1d78[8]);
var out = obj.OpenTextFile(_0x1d78[9], 2, true, 0);
out.WriteLine(_0x1d78[10]);
out.Close();
var fle = obj.GetFile(_0x1d78[9]);
fle[_0x1d78[11]] = 2
} catch (err) {
WScript.Echo(_0x1d78[12])
}
```

b. Fiecare **\x$$** este o secventa escape hexazecimala care reprezinta un caracter ASCII. Acestea sunt folosite pentru a ascunde textul original. Daca decodam fiecare valoare si le concatenam, obtinem exact aceleasi mesaje ca in sample2.js, doar ca stocate intr-un array si accesate prin indecsi.

c. Diferenta consta in metoda de ofuscare: sample2.js foloseste **eval()** impreuna cu o functie automat generata (obfuscator tip Packer sau JJencode), in timp ce sample3.js foloseste **coduri hexazecimale** (\x$$) pentru a ascunde textul.
Din punct de vedere al comportamentului (analiza dinamica), rezultatul este acelasi: aceleasi pop-up-uri si actiunea asupra fisierului fmi.txt. Totusi, sintaxa este usor diferita: in sample3.js stringurile sunt scrise cu ', iar setarea atributului fisierului este scrisa explicit cu fle['attributes'] = 2 in loc de fle.attributes = 2.

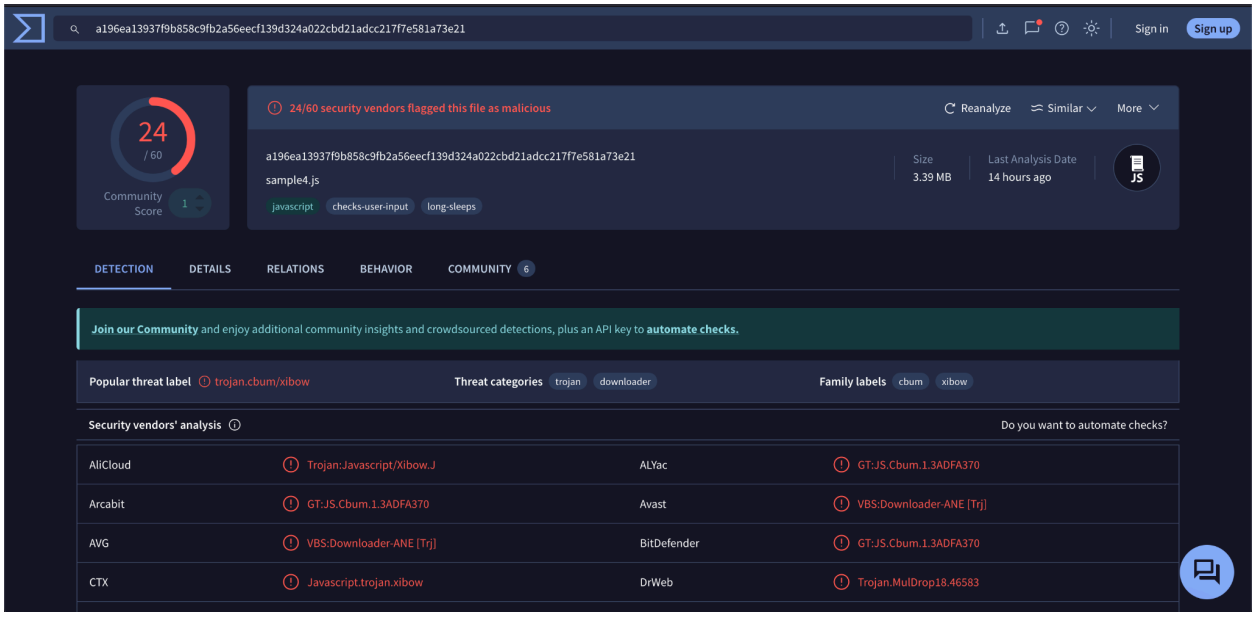
4. sample4.js

a. Codul afiseaza un message box cu “Hello!”

b. -

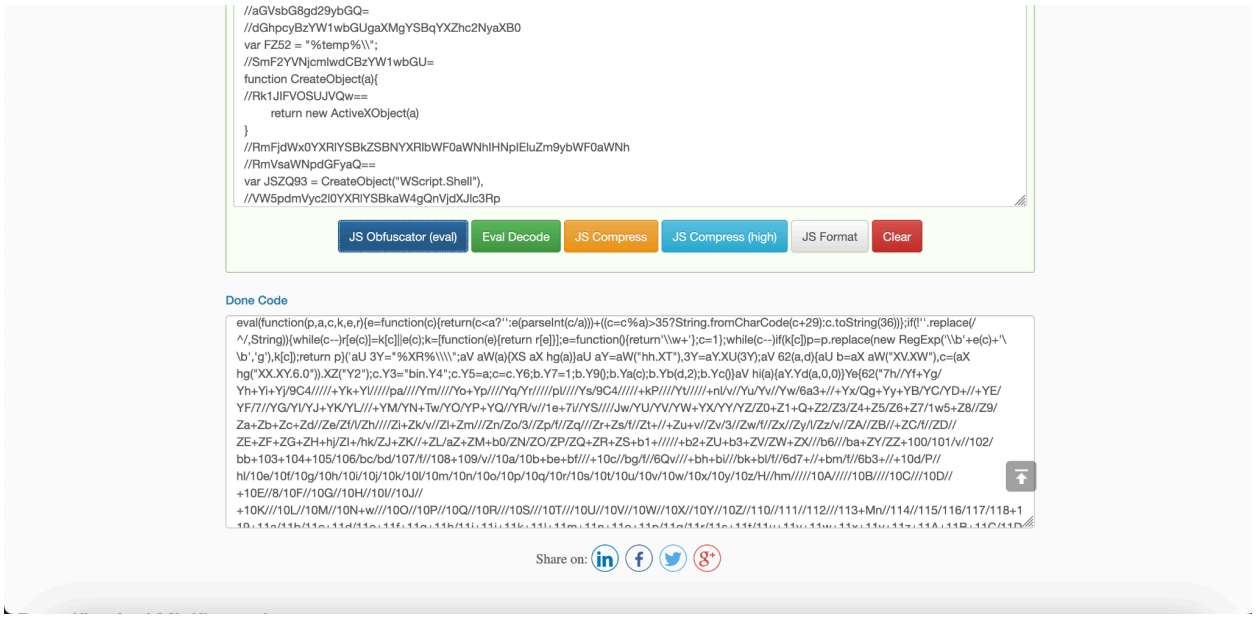
c. -

d. Utilizare **VirusTotal**

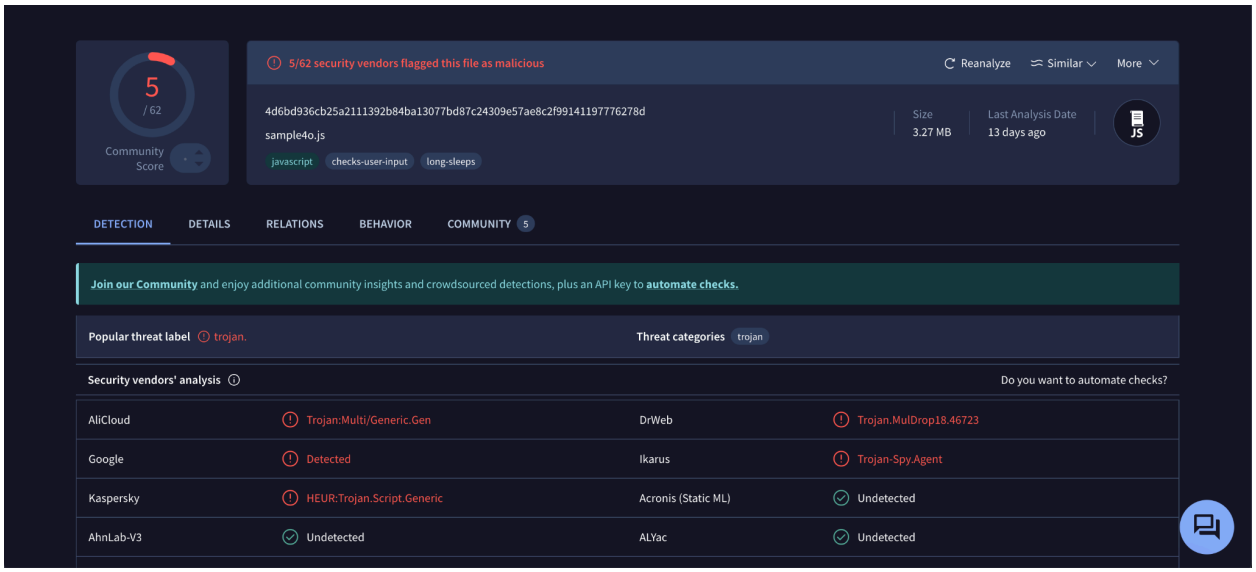


24 din 60 de “security vendors” au dat flag la acest fisier deci s-ar putea sa fie malware.

e. Utilizare **BeautifyConverter JavaScript**



Fisierul dupa obfuscare:



Daca il obfuscam, fisierul devine mai greu de citit si astfel mai greu de detectat daca este malware.